

CPS635



Description

The CPS635 Collimated Laser Diode Module has a typical center wavelength close to 635 nm and produces an output beam that has an elliptical beam shape.

Our CPS Series of Compact Laser Diode Modules are engineered to withstand large temperature variations. The module housing is precision machined by a lathe after assembly to maintain the parallelism (axis deviation) between the housing and output beam. The CPS635 features an Ø11 mm housing with an 18" (457 mm) strain relief cable and is compatible with our AD11F, AD11BA, AD11NT, KAD11F, and KAD11NT mounting adapters and LDS5 power supply. Each CPS635 laser diode module is shipped with a test datasheet that includes the lasing spectrum.

Direct viewing of laser diode emission may cause eye damage. Extreme care must be taken to prevent the beam from being viewed directly or through external optics or mirrors.

Specifications

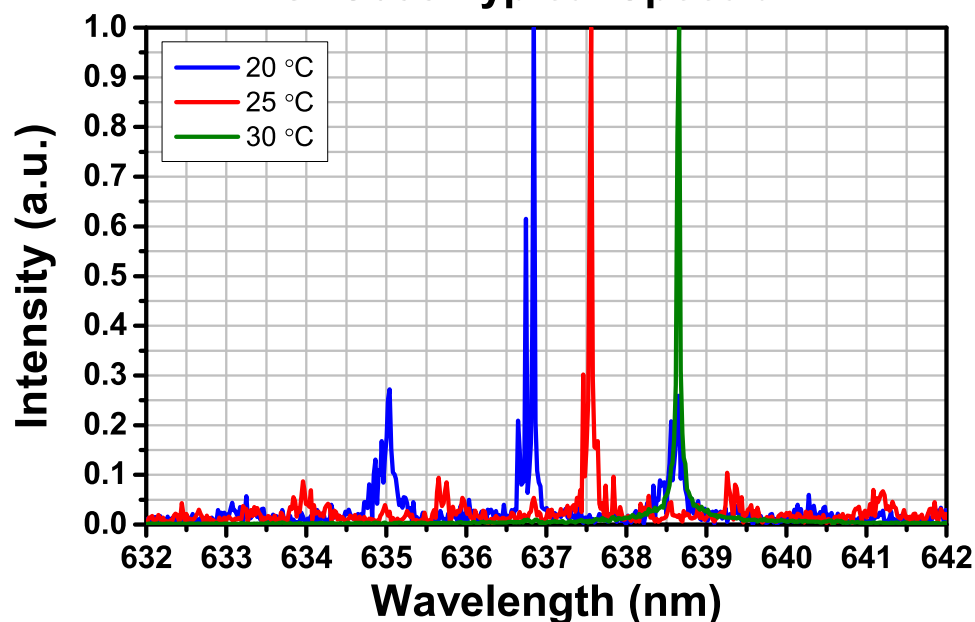
General Specifications	
Housing Material	Aluminum
Housing Dimensions	Ø11.0 mm x 40.0 mm
Beam Shape	Elliptical, 1.0 mm x 4.5 mm
Operating Temperature	-10 to 50 °C
Storage Temperature	-30 to 70 °C
Operating Voltage	4.9 V to 5.2 V
Laser Safety Rating	Class 3R

Optical and Electrical Specifications			
	Min	Typical	Max
Wavelength	-	635 nm	640 nm
Power	4.0 mW	4.5 mW	5.0 mW
Polarization Extinction Ratio	-	20 dB	-
Power Stability (8 Hours)	-	-	2%
Power Stability (1 Minute)	-	-	1%
Axis Deviation	-	-	5 mrad
Beam Divergence	-	-	1.5 mrad
Operating Current	-	48 mA	70 mA

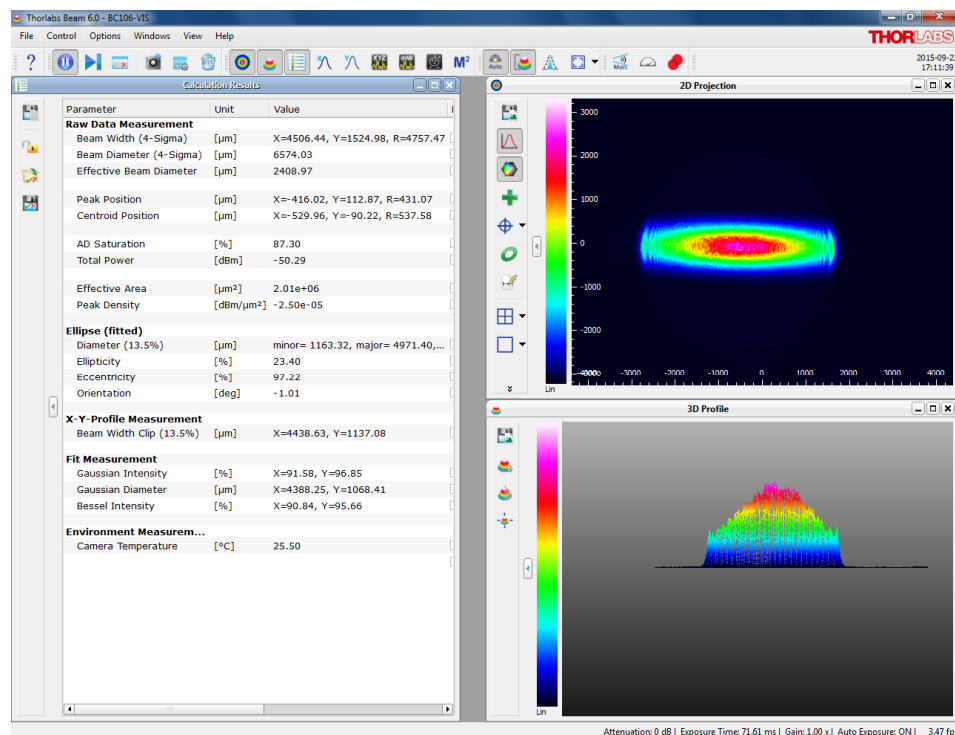


Typical Performance Plots

CPS635 Typical Spectrum

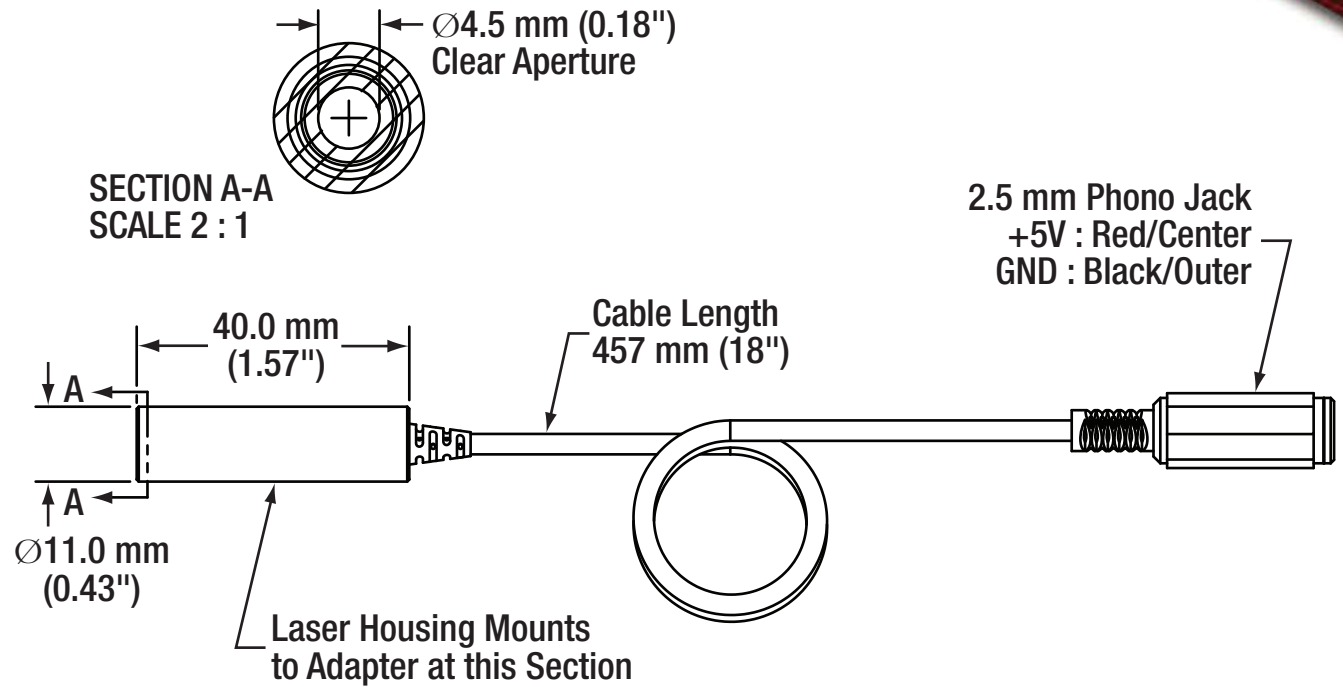


Spectrum of the CPS635 Diode Module taken at 20 °C, 25 °C, and 30 °C. The measurement was taken using Thorlabs OSA201 Spectrum Analyzer, which has a resolution of 7.5 GHz (0.25 cm^{-1}). This data is typical and will vary for each module.



This typical beam profile of a CPS635 was obtained using Thorlabs' BC106-VIS Beam Profiler.

Drawings



October 13, 2015

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